QP CODE: 24900247

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

FIRST SEMESTER MGU- UGP (HONOURS) REGULAR EXAMINATION NOVEMBER 2024

FIRST SEMESTER

MG1DSCSTA100 – FUNDAMENTALS OF STATISTICS AND DATA VISUALISATION

Duration: 1.5 hours **Maximum marks: 50**

SCHEME OF VALUATION

Part - A

Short Answer Questions

Answer any seven questions. Each question carries two marks

1. Definition or drawing the shape of histogram - 2 marks
2. Defining both absolute measure of dispersion and relative measure of dispersion – 1 + 1 marks or writing an example each for both - 1+1 marks
3. Any one example - 2 marks. (If only definition of stratified sampling - 1 mark)
4. Definition of both discrete and continuous variable - 1+1 marks or one example each for discrete variable and continuous variable - 1 +1 marks
5. P(AB) = P(A) P(B) - 1 mark; P(AB) = 0.5 X 0.6 = 0.3 - 1 mark (Even if no formula and answer is correct, give 2 marks)
6. P(A|B) = P(AB)/ P(B) - 1 mark; Answer = (1/8) / (1/4) = 0.5 - 1 mark
7. Definition - 1 mark (Defining the concept or drawing the shape of a positively skewed data) ; Relation: mean > median > mode - 1 mark
8. Defining both correlation and regression - 1 +1 marks; (defining the concepts or writing Pearson formula for correlation / Spearman's rank correlation formula and formulae for regression equations, give 2 marks)
9. Formula: r = $\sqrt{bd}$ - 1 mark; r = 0.6 - 1 mark;
10. Equation: x - $\bar{x}$ = d (y - $\bar{y}$) where d is the x on y regression coefficient - 1 mark; x – 3.5 = 1.136 (y - 2.8) - 1 mark

Part B

Short Essay Questions

Answer any four questions. Each question carries six marks

11. Formula for second, third central moments and formula for skewness - ½ +1/2 + 1 marks. Table or calculations – 2 marks; $m_3 = 1.597$, $m_2 = 23.471$, skewness = 0.00019 - ½ + ½ + 1 marks . (Slight variations in answers due to decimal approximations are acceptable).
12. Assuming the two equations as y on x and x on y respectively - 2 marks; r = $\sqrt{\frac{3}{4} x \frac{3}{4}}$ = .75 < 1 - 2 marks; assumption is right . first equation - y on x and second equation - x on y - 2 marks
13. P(AUB) = P(A) + P(B) – P(AB) - 1 mark; P(AB) = 0.5 + p -0.8 = p – 0.3 - 1 mark; (1) A and B are mutually exclusive, then P(AB) = 0 and hence p = 0.3 - 2 marks; (2) A and B are independent , P(AB) = P(A) . P(B) , p – 0.3 = 0.5x p, then p = 0.6 - 2 marks
14. Arrange values in ascending or descending order -1 mark; Q1= (n +1)/4 th = 2.25^{th} = 2.725 - 2 marks; Q2 = 3.085 - 1 mark; Q3 = $3(n+1)/4^{th} = 6.75^{th}$ = 3.2625 - 2 marks (In the question paper, 2.63 and 2.89 are printed as 2,63 and 2,89 respectively. If the problem is solved by considering these data values as 2, 63, 2, 89, then also give full marks)
15. Arrange values in ascending or descending order - 1 mark; median = 1500 – 1 mark; table or calculations - 2 marks; formula for mean deviation – 1 mark; Answer = 338.888 - 1 mark
16. Writing first equation as y = $\frac{2}{5}$ x - $\frac{2}{5}$ - 2 marks; writing second equation as x = $\frac{4}{5}$ y - 4 - 2 marks; r = $\sqrt{b.d}$ and getting r = 0.5656 - 2 marks

Part C . Essay Questions

Answer any one question. Each question carries 12 marks

17. Defining both probability sampling techniques and non-probability sampling techniques, by mentioning names of 3 techniques each - 3 +3 marks; Giving real life examples for any of the two methods - 3 + 3 marks
18. Writing the two regression equations - 2 marks; $\Sigma x = 42$, $\Sigma y = 78$, $\Sigma xy = 595$, $\Sigma x^2 = 340$, $\Sigma y^2 = 1070$ - ½ mark each; getting $\bar{x}$ = 7 and $\bar{y}$ = 13 – 1 mark; co variance = 8.166 – ½ mark; getting y on x regression coefficient as 1.065 and x on y regression coefficient as 0.874 - 2 marks; writing y on x equation as : y – 13 = 1.065 (x – 7) and x on y regression equation as x – 7 = 0.874 (y – 13) - 2 marks; Estimated value of X when Y = 20, is 13. 118 - 1 mark; Estimated value of Y when X = 20, is 26.845 - 1 mark

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